

Department of Electrical Engineering
Dr. B. C. Roy Engineering College, Durgapur

Safe Operating Area (SOA)

Power Semiconductor Devices: FBSOA, RBSOA and MOSFET SOA
Datasheet-Based Interpretation, Numerical Checks and RCD Appendix

Devices covered:

2N3055 (NPN BJT, TO-3) → **IRF540N** / **IRF540NPbF** (N-Ch MOSFET, TO-220)

 **Kingsuk Majumdar, Ph.D. (Engg)**
 Assistant Professor (Grade II), Dept. of EE, BCREC
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Note

The note uses the SOA interpretation associated with Mohan et al. [1] and the manufacturer PDF datasheets for the 2N3055 [2] and IRF540NPbF [3]. The two supplied SOA figures are used as the principal graphical references.

Main text: SOA only.

RCD turn-off snubber treatment is intentionally placed in Appendix A.

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1 Introduction: What Is the Safe Operating Area?

The Safe Operating Area (SOA) is the envelope of voltage-current combinations within which a power semiconductor can operate without exceeding the limits specified by the manufacturer. Following the conventional log-log SOA plot, current is placed on the vertical axis and device voltage on the horizontal axis [1].

Note

Axis convention: the Y-axis is I_C or I_D and the X-axis is V_{CE} or V_{DS} . A maximum-current limit appears as a horizontal boundary, while a maximum-voltage limit appears as a vertical boundary [1].

For a BJT, the safe region depends on the bias condition during the relevant interval. The two important cases are **FBSOA** during forward-bias operation and **RBSOA** during reverse-biased turn-off. A MOSFET is normally specified by one SOA family of curves rather than a separate BRSOA/RBSOA classification [3].

2 BJT SOA: FBSOA and RBSOA

2.1 Forward-Bias Safe Operating Area (FBSOA)

Definition

FBSOA is the allowable combination of V_{CE} and I_C while the BJT base-emitter junction remains forward biased. It is the relevant SOA for normal on-state, linear-region or pulsed conduction with the specified base drive.

The FBSOA is bounded by current capability, thermal power dissipation, secondary breakdown and the applicable collector-emitter voltage limit.

1. Maximum collector current

$$I_C \leq I_{C(\max)}. \quad (1)$$

For the 2N3055, the continuous current rating is 15 A under the stated datasheet conditions [2].

2. Thermal / constant-power limit

For a DC operating point at case temperature T_C ,

$$V_{CE}I_C \leq P_D, \quad P_D \approx \frac{T_{J(\max)} - T_C}{R_{\theta JC}}. \quad (2)$$

For short pulses, the relevant thermal quantity is transient thermal impedance $Z_{\theta JC}(t_p)$, so the safe power can rise above the DC value. The manufacturer therefore provides distinct pulse-duration boundaries [2].

3. Secondary breakdown

Secondary breakdown is a distinctive limitation of power BJTs. At sufficiently high V_{CE} and I_C , current crowding and localized thermal feedback can produce a much more restrictive boundary than the simple power hyperbola. The real device SOA curve, rather than an invented straight-line model, should be used for design [1], [2].

4. Collector-emitter voltage ceiling

$$V_{CE} \leq BV_{CEO}. \quad (3)$$

2.2 Manufacturer FBSOA and RBSOA Figure

Figure 1 is the supplied composite figure for the 2N3055. The left panel is the forward-bias SOA and the right panel is the reverse-bias SOA. It is intentionally reproduced here because it provides the most direct visual interpretation of the current, voltage, thermal and secondary-breakdown boundaries for this specific device [2].

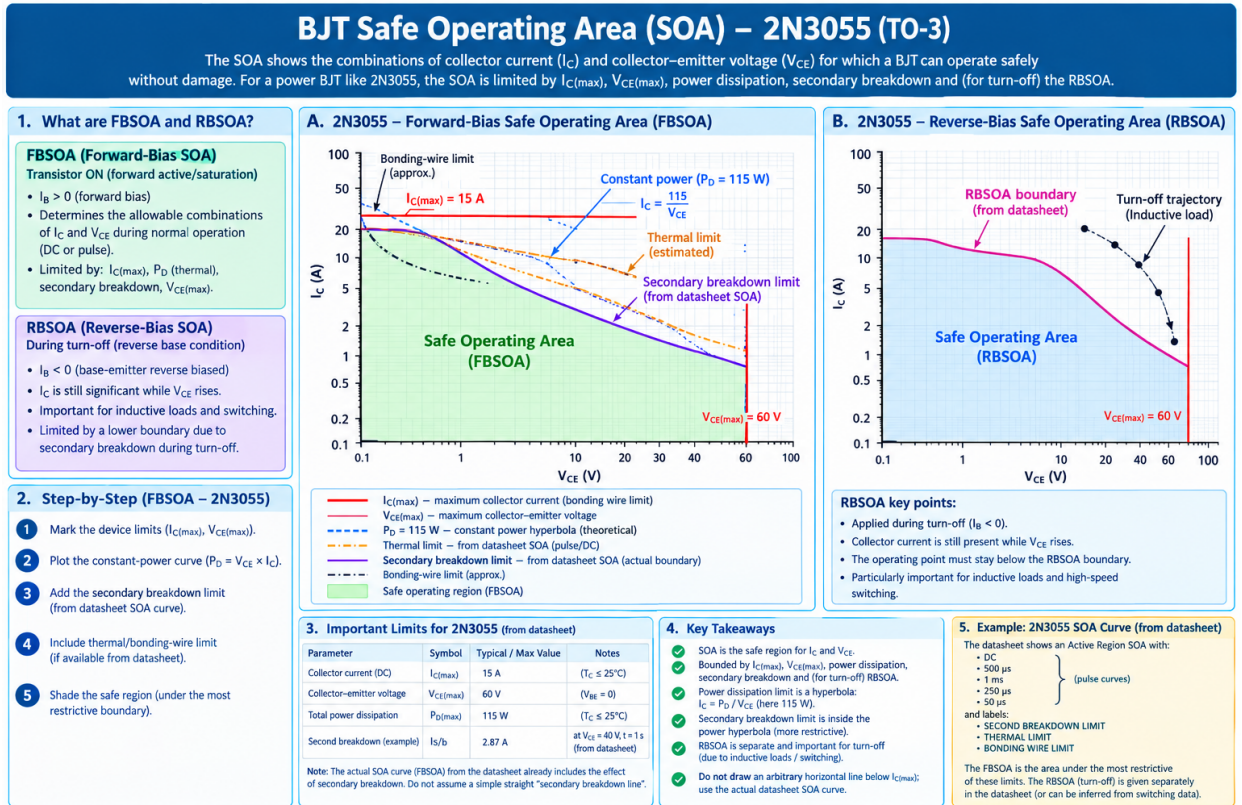


Figure 1: 2N3055 SOA figure supplied with these notes. The left panel shows FBSOA with current, thermal, secondary-breakdown and voltage constraints; the right panel shows the smaller RBSOA envelope relevant to turn-off. The supplied figure is used here as the device-specific graphical reference [2].

2.3 Reverse-Bias Safe Operating Area (RBSOA)

Definition

RBSOA applies during turn-off when a reverse base current is used to remove stored charge. During the transient, I_C can remain significant while V_{CE} rises. The allowable trajectory must remain inside the RBSOA boundary.

Compared with FBSOA, the turn-off condition is more sensitive because stored charge is being removed while the collector voltage is increasing. This can concentrate current in a smaller region of the die and makes the secondary- breakdown boundary especially important.

A practical SOA check therefore follows the switching trajectory in the (V_{CE}, I_C) plane rather than checking only the initial and final points. The dashed turn-off trajectory shown in the supplied RBSOA panel should be interpreted as a qualitative illustration; the actual converter trajectory must be obtained from the switching waveform or a validated device model.

Warning

For a BJT power switch, it is not sufficient to prove that the on-state point lies within FBSOA. The complete turn-off trajectory must also remain within the manufacturer RBSOA envelope [2].

3 MOSFET Safe Operating Area

Definition

For the IRF540N, the manufacturer provides one SOA family of curves. The plot contains the current-limited region, voltage limit, thermal/constant-power region and pulse-duration dependence. There is no separate RBSOA curve of the BJT type [3].

3.1 Main SOA boundaries

1. Current limit

$$I_D \leq I_{D(\max)}. \quad (4)$$

2. Low- V_{DS} / $R_{DS(on)}$ limit

In the ohmic region,

$$V_{DS} \approx I_D R_{DS(on)}. \quad (5)$$

As junction temperature increases, $R_{DS(on)}$ also increases, which changes the practical high-current boundary.

3. Thermal / power limit

$$V_{DS} I_D \leq P_D, \quad (6)$$

with the usable power depending on junction temperature, case temperature, heat sinking and pulse duration.

4. Drain-source breakdown voltage

$$V_{DS} \leq BV_{DSS}. \quad (7)$$

3.2 Manufacturer SOA Figure

Figure 2 shows the supplied IRF540N SOA plot with the 100- μ s, 1-ms, 10-ms and single-pulse curves. The key point is that pulse duration changes the permissible I_D at a given V_{DS} , so a single constant-power estimate must not replace the datasheet SOA curve [3].

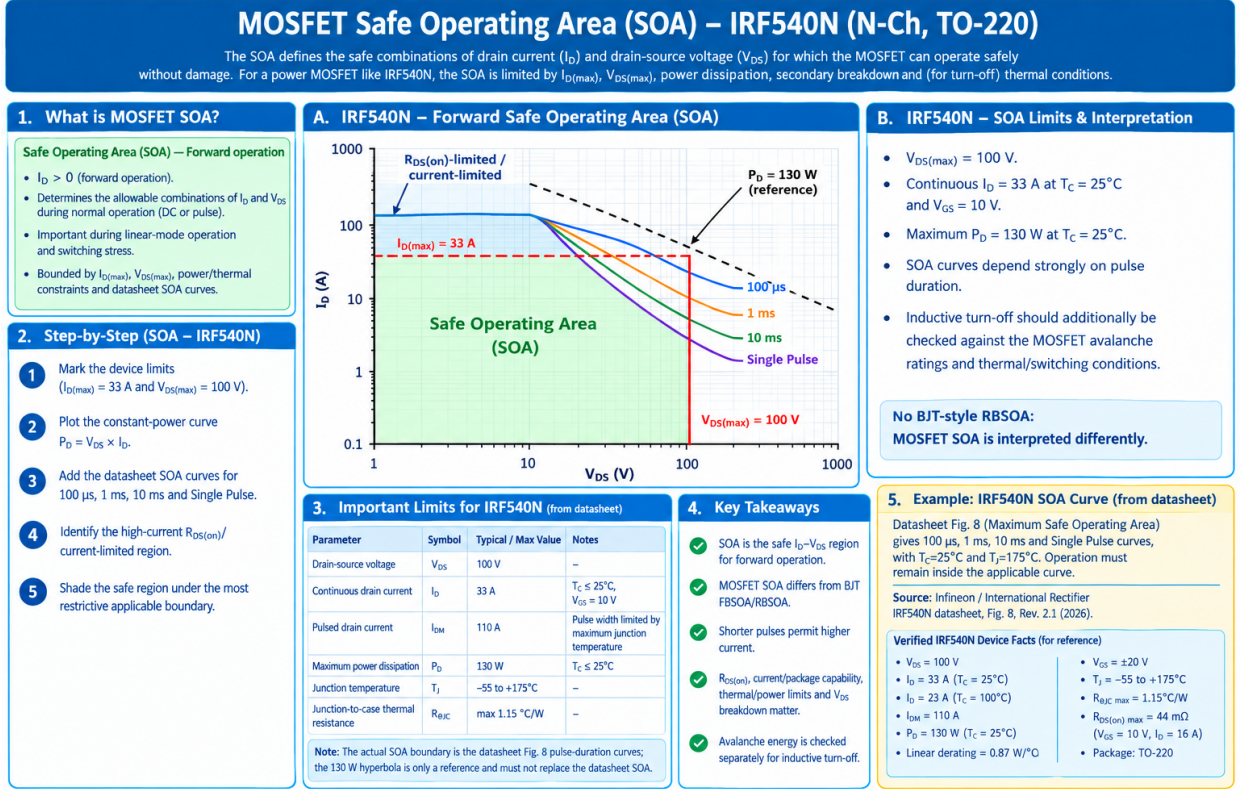


Figure 2: IRF540N SOA figure supplied with these notes. The plot shows the $R_{DS(on)}$ /current-limited region, the 100-V voltage ceiling, a reference P_D hyperbola, and pulse-duration curves. The applicable manufacturer SOA curve is the controlling limit [3].

4 Numerical SOA Check: 2N3055

4.1 Representative operating point

Use $V_{CC} = 50$ V, $I_C = 4$ A, an inductive load and $T_C = 25^\circ\text{C}$ for the illustrative check.

4.2 FBSOA check

Current: 4 A < 15 A, so the current rating is satisfied.

On-state voltage: for the stated example, $V_{CE} \approx 2$ V < 60 V.

Table 1: Key 2N3055 parameters from the onsemi datasheet [2]

Parameter	Symbol	Value	Unit
Collector-emitter voltage	BV_{CEO}	60	V
Collector-base voltage	BVC_{BO}	100	V
Continuous collector current	I_C	15	A
Power dissipation at $T_C = 25^\circ\text{C}$	P_D	115	W
Derating above 25°C	–	0.657	W/ $^\circ\text{C}$
$R_{\theta JC}$	–	1.52	$^\circ\text{C}/\text{W}$
Junction/storage temperature	T_J, T_{stg}	–65 to +200	$^\circ\text{C}$
Second-breakdown point	$I_{s/b}$	2.87 at $V_{CE} = 40\text{ V}, t = 1\text{ s}$	A

DC conduction power:

$$P_{on} = V_{CE}I_C \approx 2 \times 4 = 8\text{ W} < 115\text{ W}. \quad (8)$$

Thus the selected on-state point is well inside the basic DC current, voltage and power ratings. The full pulsed operating point must still be checked against the manufacturer FBSOA curve in Fig. 1.

4.3 RBSOA check at turn-off

At turn-off, V_{CE} can rise toward V_{CC} before I_C has fallen to zero. The datasheet gives $I_{s/b} = 2.87\text{ A}$ at $V_{CE} = 40\text{ V}$ and $t = 1\text{ s}$ under the stated test condition [2].

At the illustrative current,

$$I_C = 4\text{ A} > 2.87\text{ A} \quad \text{at} \quad V_{CE} = 40\text{ V}. \quad (9)$$

Warning

This comparison indicates that the illustrative turn-off condition is above the listed second-breakdown test point. It should not be interpreted as a complete replacement for the device-specific RBSOA curve: pulse width, base drive and waveform trajectory matter. The supplied RBSOA plot is therefore the primary turn-off reference for the device.

5 Numerical SOA Check: IRF540N / IRF540NPbF

5.1 Illustrative operating point

Take $I_D = 10\text{ A}$ and $V_{DS} = 80\text{ V}$ for a simple margin check. The point is below the headline ratings $I_D = 33\text{ A}$ and $V_{DS} = 100\text{ V}$, while the exact safe combination must still be read from the pulse-duration curve in Fig. 2.

$$P = V_{DS}I_D = 80 \times 10 = 800\text{ W}. \quad (10)$$

Table 2: Key IRF540N / IRF540NPbF parameters from the manufacturer PDF [3]

Parameter	Symbol	Value	Unit
Drain-source breakdown voltage	BV_{DSS}	100	V
Continuous drain current at $T_C = 25^\circ\text{C}$	I_D	33	A
Pulsed drain current	I_{DM}	110	A
$R_{DS(on)}$ max at $V_{GS} = 10\text{ V}$	$R_{DS(on)}$	44	m Ω
Power dissipation at $T_C = 25^\circ\text{C}$	P_D	130	W
$R_{\theta JC}$ max	–	1.15	$^\circ\text{C/W}$
$R_{\theta CS}$ typ.	–	0.50	$^\circ\text{C/W}$
Junction temperature	T_J	–55 to +175	$^\circ\text{C}$

⚠ Warning

The 800 W instantaneous $V_{DS}I_D$ product is far above the 130-W DC power-dissipation rating, so the simple 80-V, 10-A point is **not a safe DC operating point**. It can only be discussed, if at all, as a very short pulse with the appropriate manufacturer SOA curve. This is precisely why the SOA graph, rather than a rectangular combination of individual absolute maximum ratings, must be used [3].

5.2 Correct interpretation

For the IRF540N, an SOA check should proceed by locating the actual (V_{DS}, I_D) trajectory on the manufacturer curve for the required pulse duration. The $R_{DS(on)}$ region controls at low V_{DS} and high current, while the thermal and breakdown limits become increasingly important as V_{DS} rises [3].

6 BJT versus MOSFET: SOA Interpretation

Conclusion

SOA analysis is fundamentally a trajectory check. For the 2N3055, the forward bias SOA governs normal conduction while the smaller RBSOA governs the turn-off transient. The manufacturer figure is the authoritative device-specific boundary for both cases [2].

For the IRF540N, the manufacturer provides a single SOA family with distinct pulse-duration curves. A voltage-current point that satisfies separate absolute maximum ratings can still be unsafe when the product $V_{DS}I_D$ is too large for the relevant pulse duration [3].

The main note therefore keeps the discussion focused on SOA itself. The requested RCD turn-off snubber design is separated into Appendix A so that the SOA concepts, the device-specific curves, and the mitigation network remain clearly distinguished.

Table 3: SOA comparison of the 2N3055 and IRF540N

Property	2N3055 BJT	IRF540N MOSFET
SOA form	FBSOA + RBSOA	Single SOA family
Current limitation	Package/device current limit	Current / $R_{DS(on)}$ region
Thermal limitation	DC and pulse thermal limits	DC and pulse thermal limits
Secondary breakdown	Important SOA constraint	No BJT-type secondary-breakdown boundary
Turn-off check	RBSOA trajectory required	Read the applicable SOA curve
Pulse duration	Strong effect on SOA	Strong effect on SOA
Main design lesson	Check both on-state and turn-off envelopes	Do not infer SOA from absolute maximum ratings alone

A RCD Turn-Off Snubber Versus RC Turn-On Snubber for a BJT

For a power BJT, the switching transitions during turn-on and turn-off can produce significant voltage and current stress. An RC turn-on snubber and an RCD turn-off snubber are designed to address different portions of the switching process.

A.1 RCD Turn-Off Snubber

An RCD turn-off snubber is primarily used to limit the collector-emitter voltage rise and voltage overshoot when the BJT is turned off, particularly when the load is inductive. During turn-off, the load current cannot change instantaneously. The inductive energy therefore tends to force the collector-emitter voltage to rise rapidly.

The RCD network provides a path for this transient energy. The capacitor temporarily absorbs the energy associated with the voltage spike, while the resistor subsequently dissipates the stored energy. The diode provides a preferred charging path for the capacitor during the turn-off transient.

The energy associated with an inductive load can be expressed as

$$E_L = \frac{1}{2}LI^2. \quad (11)$$

During the transient, part of this energy is transferred to the snubber capacitor,

$$E_C = \frac{1}{2}C_S V_C^2, \quad (12)$$

thereby limiting the peak collector-emitter voltage. The resistor then dissipates the stored energy.

The principal objectives of the RCD turn-off snubber are therefore to

- limit the peak collector-emitter voltage $V_{CE,peak}$,

- reduce the rate of rise of collector-emitter voltage, dv_{CE}/dt ,
- suppress voltage overshoot and ringing, and
- reduce the turn-off voltage stress on the BJT.

The RCD turn-off snubber is particularly important in inductive switching because the resulting (V_{CE}, I_C) trajectory must remain within the reverse-bias safe operating area (RBSOA) of the transistor.

A.2 RC Turn-On Snubber

An RC turn-on snubber is intended primarily to control the BJT turn-on transition. During turn-on, the collector-emitter voltage decreases while the collector current increases. Without suitable control, this transition can result in a large voltage/current overlap and consequently high instantaneous switching power,

$$p_{sw}(t) = V_{CE}(t)I_C(t). \quad (13)$$

The snubber capacitor limits the rate of change of the device voltage according to

$$i_C = C_S \frac{dv}{dt}. \quad (14)$$

Thus, an appropriate choice of the snubber capacitance can reduce the dv/dt during turn-on. The resistor limits the transient current and provides a path for the discharge of the capacitor.

The principal objectives of an RC turn-on snubber are therefore to

- reduce the turn-on dv/dt ,
- reduce voltage/current overlap during turn-on,
- limit turn-on switching stress, and
- suppress high-frequency transient ringing.

A.3 Comparison of RCD Turn-Off and RC Turn-On Snubbers

The two snubber configurations address different switching transitions and should not be considered interchangeable. Their main differences are summarized in Table 4.

A.4 Relation to BJT Safe Operating Area

For an inductive load, the turn-off trajectory of the BJT can be represented in the V_{CE} - I_C plane. As the transistor turns off,

$$I_C \rightarrow 0, \quad V_{CE} \rightarrow V_{CE,peak}. \quad (15)$$

Without adequate clamping, the voltage may increase rapidly while appreciable collector current is still flowing. The instantaneous switching power is then

$$P(t) = V_{CE}(t)I_C(t), \quad (16)$$

Table 4: Comparison of RCD Turn-Off and RC Turn-On Snubbers for a Power BJT

Feature	RCD Turn-Off Snubber	RC Turn-On Snubber
Main purpose	Limit turn-off voltage overshoot	Control turn-on voltage transition
BJT transition	Turn-off	Turn-on
Primary stress	High V_{CE} during current interruption	High dv/dt and voltage/current overlap
Typical application	Inductive switching	Controlled turn-on switching
Capacitor function	Absorbs transient turn-off energy	Controls the rate of voltage change
Resistor function	Dissipates stored capacitor energy	Limits transient current and dissipates energy
Diode	Required for the preferred unidirectional charging path in the RCD clamp	Generally not required
SOA relevance	Directly related to turn-off/RBSOA stress	Primarily related to turn-on switching stress

which can become large enough to violate the transistor's reverse-bias safe operating area.

The RCD snubber reduces the severity of this trajectory by limiting the collector-emitter voltage excursion. Consequently, the device operating trajectory can be kept within the specified RBSOA, provided that the snubber is properly designed.

The RC turn-on snubber, on the other hand, primarily modifies the turn-on trajectory by limiting the rate at which V_{CE} falls. It is therefore mainly associated with turn-on switching loss and transient control rather than the reverse-bias turn-off SOA.

RC snubber: primarily controls BJT turn-on transition

RCD snubber: primarily clamps BJT turn-off voltage

(17)

Thus, for the SOA analysis of a power BJT such as the 2N3055, the RCD turn-off snubber is particularly relevant when evaluating the turn-off trajectory against the RBSOA boundary.

B RCD Turn-Off Snubber for the 2N3055[4]

This appendix gives the requested illustrative RCD turn-off snubber calculation for the representative 2N3055 inductive-load case. It is intentionally placed outside the main SOA section. The purpose of the network is to reshape the turn-off trajectory; the final design must still be verified against the actual RBSOA curve and measured switching waveform [1], [2].

B.1 Topology and operating principle

The snubber branch consists of a diode D_s in series with the parallel combination $R_s \parallel C_s$, connected between the collector node and the emitter rail. During turn-off, D_s conducts and C_s absorbs transient energy; during the following interval, R_s dissipates the stored energy as heat.

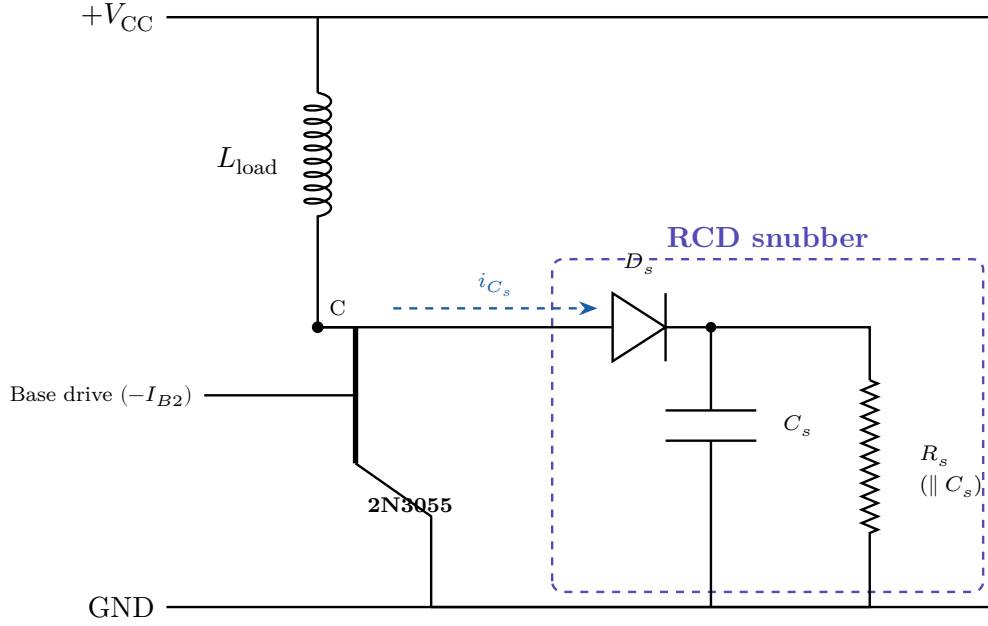


Figure 3: Illustrative RCD turn-off snubber topology used in Appendix A.

B.2 Design example

Use $V_{CC} = 50\text{ V}$, $I_C = 4\text{ A}$, $t_f = 3\text{ }\mu\text{s}$, $f_s = 10\text{ kHz}$, and a target $V_{CE}^{\max} = 55\text{ V}$.

Step 1: Capacitor C_s

Assuming an approximately linear current fall,

$$V_{CE}^{\max} \approx V_{CC} + \frac{I_C t_f}{2C_s}. \quad (18)$$

Therefore,

$$C_s = \frac{I_C t_f}{2(V_{CE}^{\max} - V_{CC})} = \frac{4 \times 3 \times 10^{-6}}{2(55 - 50)} = 1.2\text{ }\mu\text{F}. \quad (19)$$

Choose $C_s = 1.5\text{ }\mu\text{F}$ as the next practical value, with suitable voltage and pulse-current margin.

Step 2: Resistor R_s

For a 50% duty-ratio example, $T_{on} = 1/(2f_s) = 50\text{ }\mu\text{s}$. Requiring about five time constants gives

$$5R_s C_s \leq T_{on}, \quad R_s \leq \frac{50 \times 10^{-6}}{5 \times 1.5 \times 10^{-6}} = 6.67\text{ }\Omega. \quad (20)$$

Choose $R_s = 5.6\text{ }\Omega$ as an illustrative standard value.

Step 3: Resistor power

$$E_{C_s} = \frac{1}{2} C_s V_{CC}^2 = \frac{1}{2} (1.5 \times 10^{-6}) (50)^2 = 1.875\text{ mJ}. \quad (21)$$

At $f_s = 10\text{ kHz}$,

$$P_{R_s} = E_{C_s} f_s = 18.75\text{ W}. \quad (22)$$

An illustrative resistor rating is therefore $\geq 25\text{ W}$, with final thermal verification on the actual assembly.

Step 4: Snubber diode

The diode must have adequate repetitive reverse-voltage, forward-current, recovery-time and thermal ratings. The illustrative selection from the original calculation is MUR410; its suitability must be verified against the final switching waveform and operating temperature.

Table 5: Illustrative RCD snubber values for the 2N3055 example

Component	Value	Illustrative rating	Function
C_s	1.5 μF	$\geq 100\text{ V}$	Limits dV_{CE}/dt
R_s	5.6 Ω	$\geq 25\text{ W}$	Dissipates stored energy
D_s	MUR410	Verify V_R , I_F , t_{rr}	Routes turn-off current

Note

The values in Appendix A are an illustrative engineering calculation, not a universal snubber recipe. Stray inductance, actual dI_C/dt , gate/base drive, duty cycle and the measured overshoot can materially change the required values.

References

References

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